Topic Modelling with BERT

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Session 1

Joy Lan

Course plan

W1: Lead by **Joy Lan**

What is topic modelling
What is BERT
How does BERT work

W2: Lead by **Aybuke Atalay**

Interpretation
Visualisation
Fine-tuning

Billions of texts. What are they saying?

Every day we generate more text than any person can read.

Journals, policy documents, reports, news, social media…

We want to (quickly) discover what people are talking about?

Where do you start?
How can we make sense of the text?

Know your data, know your goal

Q1. What do you know about your data?

- Metadata: source, timeframe, language, how it was collected?..
- Content: general topic, coherence, length, etc.

Q2. What is your analytical goal?

- Exploring patterns in a collection of documents
- Classify documents into categories
- Analysing meaning, framing and discourse in texts

Supervised & Unsupervised learning in NLP

Supervised learning such as text classification, can be used to:

- Train a model that can later detect or predict a “defined label”.
- Validate and test coding scheme.

You’ll need:

- A defined label set
- **Labelled** documents
- Model configurations

You’ll get:

- A trained model that can assign predicted label(s) to new data
- Evaluation metrics that help you assess the model performance

Supervised & Unsupervised learning in NLP

Unsupervised learning , such as topic modelling can be used to:

- Train a model that can later detect or predict a “defined label”.
- Validate and test coding scheme.

You’ll need:

- An optional label set
- **(Unlabelled)** documents
- Model configurations

You’ll get:

- Topics/clusters described by keywords/representative texts
- Topic–document relationships

When do you use topic modelling in NLP?

Goal	Input	Output	Choice
• Detection and prediction • Validate and test coding scheme	• A defined label set • **Labelled** documents • Model configurations	• Predicted label(s) for new documents • Evaluation metrics on test data	supervised learning
• Discover themes/patterns • Generate candidate codes for qualitative analysis • Support sense-making and sampling for close reading	• A defined label set (opt.) • **Unlabelled** documents • Model configurations	• Topics/clusters described by keywords/representative texts • Topic–document relationships	unsupervised learning Topic modelling

Machine learning methods cannot replace human interpretation.

How does topic modelling work?

Topic modelling: technical workflow

a statistical method to discover topics in a set of documents

1. Cleaning
2. Tokenisation
3. Representation (turn text into numbers)
4. Train model
5. Evaluation + visualisation

BERTopic v.s LDA

Text Representation

- BoW: What words are there?
- TF-IDF: Which words matter most?
- BERT: What does the word mean here?

Text Representation - BoW

Corpus containing 2 texts:

Text 1:
I went to the **bank** to check my **interest** and a **charge** on my account, but the **current** balance looked wrong.

Text 2 :
The **current** in the **stream** created a steady **flow** of water, while the **battery** powered a **light** in the open **field** near the **bank**.

Word	T1	T2
bank	1	1
interest	1	0
charge	1	0
current	1	1
battery	0	1
light	0	1
stream	0	1
field	0	1
flow	0	1

Text Representation – Embedding

Corpus containing 2 texts:

Text 1:
I went to the **bank** to check my **interest** and a **charge** on my account, but the **current** balance looked wrong.

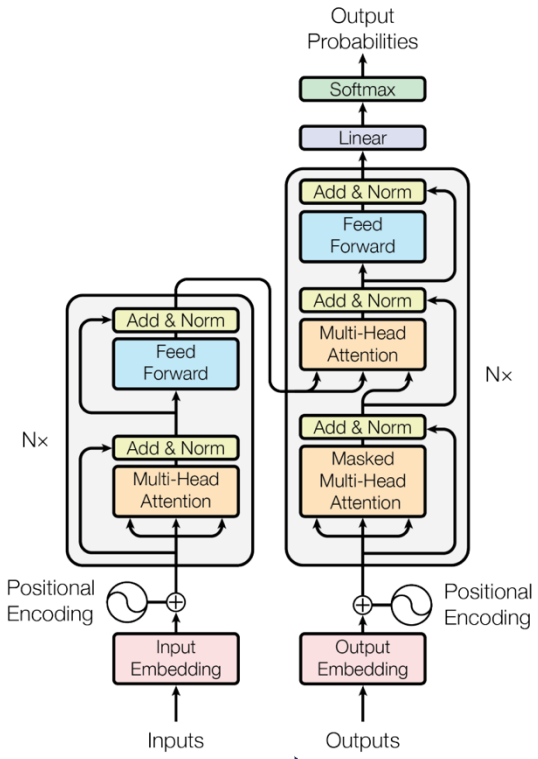
Text 2 :
The **current** in the **stream** created a steady **flow** of water, while the **battery** powered a **light** in the open **field** near the **bank**.

Word	T1	T2
bank	[0.8, 0.7, 0.1]	[0.2, 0.3, 0.6]
current	[0.7, 0.6, 0.2]	[0.3, 0.4, 0.2]
interest	[0.9, 0.8, 0.1]	—
stream	—	[0.2, 0.3, 0.9]
battery	—	[0.3, 0.5, 0.8]

BERT - Transformer models

BERT

Encoder



GPT

Decoder

BERT - Transformer models

Three elements:

- **Embeddings (BERT)** → define *meaning*
- **UMAP** → shapes *how documents cluster*
- **c-TF-IDF** → defines *how topics are described*

BERT - Transformer models

Define meanings:
Embedding + Positional Encoding + self attention

Embedding model turns each token in a piece of text corresponds to a vector in semantic space

Positional encoding generate another vector that represents the position (order) of the word in the text.

BERT - Transformer models

Define meanings:
Embedding + Positional Encoding + self attention

Self-attention is relational weighting, and the results reweighted the embedding vector.

ex. The animal didn't cross the street because it was too tired.

BERT - Transformer models

Define how documents cluster: UMAP

Embeddings are high-dimensional (each word is a vector)
UMAP reduces them, which allows easier clustering

Using UMAP, you can determine if you want to focus on local structure(more topics) or global structure (fewer topics)

BERT - Transformer models

Defines how topics are described: c-TF-IDF

Defines how importance is calculated, such as reducing frequent words across topic.

Define how topics are represented, such as using a phrase or a word

Let’s Code

Session 1

Codebook

Text cleaning

```
for t in en_data["text"]:
    cleaned = strip_emoji(t)                # 1. remove emojis
    cleaned = strip_all_entities(cleaned)   # 2. remove URLs, mentions, punctuation
    cleaned = clean_hashtags(cleaned)       # 3. clean hashtag patterns
    cleaned = filter_chars(cleaned)         # 4. remove unwanted symbols ($, &)
    cleaned = remove_mult_spaces(cleaned)   # 5. collapse multiple spaces
    clean_texts.append(cleaned)
```

Codebook

Embedding model

```
topic_model = BERTopic(embedding_model="all-mpnet-base-v2",
                       language="english",
                       calculate_probabilities=True,
                       verbose=True,
                       vectorizer_model=KeyphraseCountVectorizer(max_df = int(0.5*len(docs)),
                                                                 min_df = 5, decay = 0.1, delete_min_df=5.0)
                       )
```

"all-mpnet-base-v2" is a pre-trained Sentence Transformer (based on BERT-like architecture)

It converts each document into a vector (numbers) or simply put, it turns each document into one dense vector

Codebook

UMAP

```
umap_model = UMAP(
    n_neighbors=15,
    n_components=5,
    min_dist=0.0,
    metric='cosine',
    random_state=42
)
```

n_neighbors

Low → captures **local structure**

High → captures **global structure**

n_components

Dimensions after reduction

min_dist

Low → tighter clusters

High → more spread-out clusters

random_state

Allows you to reproduce the same results

Codebook

KeyphraseVectorizers

```
topic_model = BERTopic(embedding_model="all-mpnet-base-v2",
                       language="english",
                       calculate_probabilities=True,
                       verbose=True,
                       vectorizer_model=KeyphraseCountVectorizer(max_df = int(0.5*len(docs)),
                                                                 min_df = 5, decay = 0.1, delete_min_df=5.0)
                       )
```

KeyphraseCountVectorize extract phrases that describe the topics
c-TF-IDF is automatically applied during this step

max_df = 0.5 * len(docs) → Ignore phrases appearing in **too many documents**
min_df = 5 → Only keep phrases appearing in at least 5 documents
decay = 0.1 → Reduces influence of very frequent terms over time
delete_min_df = 5.0 → Removes very rare/unstable phrases

Codebook

Calculate probabilities

```
topic_model = BERTopic(embedding_model="all-mpnet-base-v2",
                       language="english",
                       calculate_probabilities=True,
                       verbose=True,
                       vectorizer_model=KeyphraseCountVectorizer(max_df = int(0.5*len(docs)),
                                                                 min_df = 5, decay = 0.1, delete_min_df=5.0)
                       )
```

Documents can partially belong to multiple topics.

Let’s Code

Session 1

Codebook

Get topic information

```
topic_model.get_topic_info()
```

	Topic	Count	Name	Representation	Representative_Docs
0	-1	2106	-1_human rights_human_rights_amp	[human rights, human, rights, amp, standup4hum...	[we hold up a human rights mirror to governmen...
1	0	169	0_myanmar_rohingya_bangladesh_military	[myanmar, rohingya, bangladesh, military, retu...	[myanmar alongside episodic violence a campaig...
2	1	122	1_lgbti_lgbti people_love_lgbt	[lgbti, lgbti people, love, lgbt, identity, se...	[stand up for equal rights amp fair treatment ...
3	2	112	2_universal declaration_declaration_udhr70_art...	[universal declaration, declaration, udhr70, a...	[watch live celebrate the 70th anniversary of ...

Each topic is represented by a group of words
-1 refers to all outliers and should typically be ignored.

Codebook

Get topic words

```
topic_model.get_topic(topic=5)

[('climate', np.float64(0.07330344140298843)),
 ('climateaction', np.float64(0.05783934058155271)),
 ('climatechange', np.float64(0.05170335324486828)),
 ('climate change', np.float64(0.045480836287091354)),
 ('change', np.float64(0.035214848611070555)),
 ('climate action', np.float64(0.02749428536810015)),
 ('climate crisis', np.float64(0.01874932097541701)),
 ('impacts', np.float64(0.01668726185312584)),
 ('action', np.float64(0.015044765487555971)),
 ('need', np.float64(0.011944187843941691))]
```

Each topic is composed of a group of words.
Words with higher scores mean that the word are more representative of the topic, compared to the other words.

Codebook

Get representative documents

```
topic_model.get_representative_docs(5)
```

```
['if you are feeling the impacts of climate change in your daily life yo
climatechange tell us below adaptourworld climateaction',
 'why is chief at cop24 because climate change is a terrible threat to h
 'we supports the right here right now global climate alliance in callin
treat climatechange as the humanrights crisis it is learn more about rhr
```

This function returns the documents sample that best embody the semantic meaning of the topic.

Codebook

Check the topic of each document

```
topic_model.topics_[:10]
```

```
[9, -1, 22, 23, 23, 23, 9, -1, -1, -1]
```

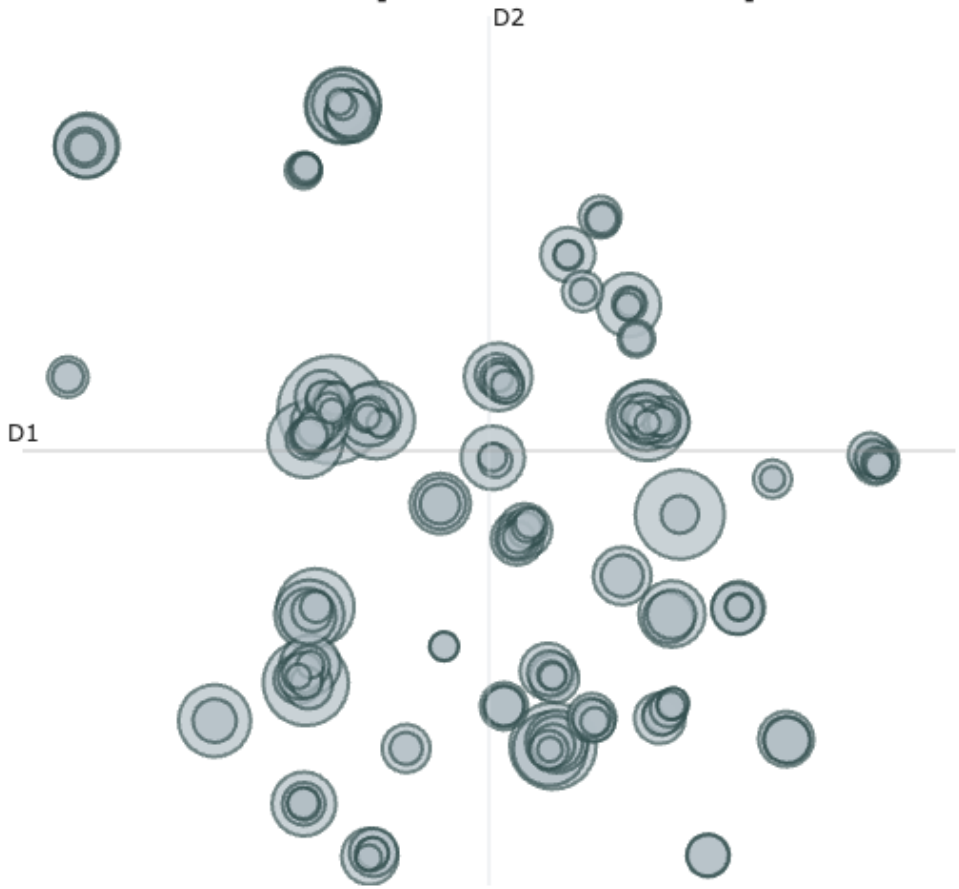
In this example,
Document 1 belongs to topic 9,
Document 2 does not have a main topic, etc.

Attributes

- There are a number of attributes that you can access after having trained your BERTopic model:

Attribute	Description
topics_	The topics that are generated for each document after training or updating the topic model.
probabilities_	The probabilities that are generated for each document if HDBSCAN is used.
topic_sizes_	The size of each topic
topic_mapper_	A class for tracking topics and their mappings anytime they are merged/reduced.
topic_representations_	The top *n* terms per topic and their respective c-TF-IDF values.
c_tf_idf_	The topic-term matrix as calculated through c-TF-IDF.
topic_labels_	The default labels for each topic.
custom_labels_	Custom labels for each topic as generated through `set_topic_labels`.
topic_embeddings_	The embeddings for each topic if `embedding_model` was used.
representative_docs_	The representative documents for each topic if HDBSCAN is used.

Intertopic Distance Map



Topic 0

Session 1 Wrap-up

Next Week:

- How can we interpret the results?
- How can we improve the model performance (that make more sense)

Additional Resources

examples of supervised learning in social studies

- Nelson, L. K., Burk, D., Knudsen, M., & McCall, L. (2021). The future of coding: A comparison of hand-coding and three types of computer-assisted text analysis methods. *Sociological Methods & Research*, *50*(1), 202-237.

- Rona-Tas, A., Cornuéjols, A., Blanchemanche, S., Duroy, A., & Martin, C. (2019). Enlisting supervised machine learning in mapping scientific uncertainty expressed in food risk analysis. *Sociological Methods & Research*, *48*(3), 608-641.

Additional Resources

examples of topic modelling

- Grubbs, A. E., Sinha, N., Garg, R., & Barber, E. L. (2023). Use of topic modeling to assess research trends in the journal Gynecologic Oncology. *Gynecologic oncology*, *172*, 41-46.

- Grajzl P, Murrell P. Using Topic-Modeling in Legal History, with an Application to Pre-Industrial English Case Law on Finance. *Law and History Review*. 2022;40(2):189-228. doi:10.1017/S0738248022000153

Thank you!